

1.1 Propositional Logic

A proposition is a declarative sentence that is either true or false, but not both.

- Quick examples

" $1 + 1 = 2$ " $\rightarrow$ proposition

"What time is it?" $\rightarrow$ not a proposition

"Read this carefully." $\rightarrow$ not a proposition

$x + 1 = 2 \rightarrow$ not a proposition

- Atomic propositions

不能再拆成更簡單命題。

cannot be broken into simpler propositions.

- Compound propositions

由較簡單命題加上邏輯運算子組成。

built from simpler propositions using logical operators

- Logical operators

- (1) Negation $\neg p$

$\neg p$ means "not p."

It flips the truth value.

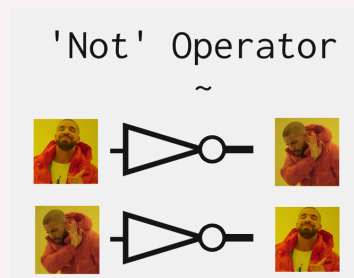
- Quick examples

p : The earth is round.

$\neg p$: The earth is not round.

*Other notations you might see are

- p , $\neg p$, $\bar{p}$, Np , and $!p$.

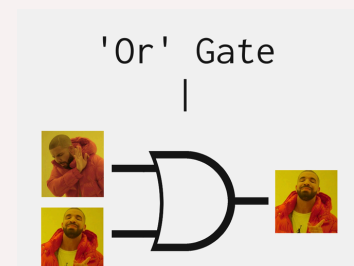


p	$\neg p$
T	F
F	T

- (2) Conjunction $p \wedge q$

$p \wedge q$ means "p and q."

It is true only when both are true.



p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

- (3) Disjunction $p \vee q$

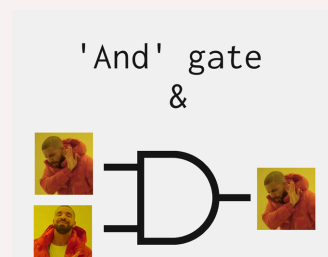
$p \vee q$ means "p or q."

It is false only when both are false.

this is usually inclusive or.

Inclusive or = one is true, or both are true.

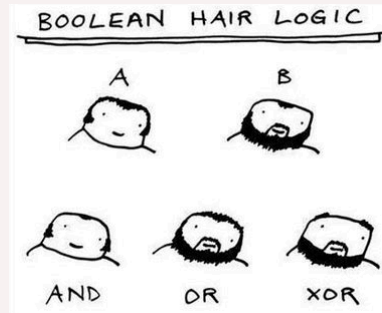
Exclusive or (XOR) = exactly one is true.



p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

(4) Exclusive or $p \oplus q$

$p \oplus q$ is true when exactly one of p and q is true.



p	q	$p \oplus q$
T	T	F
T	F	T
F	T	T
F	F	F

(5) Implication $p \rightarrow q$

$p \rightarrow q$ means "if p , then q ."

It is false only when p is true and q is false.

p is the hypothesis.

q is the conclusion.

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

(6) Biconditional $p \leftrightarrow q$

$p \leftrightarrow q$ means " p if and only if q ."

It means both $p \rightarrow q$ and $q \rightarrow p$ are true.

Operator	Priority
$\neg$	1
$\wedge$	2
$\vee$	3
$\rightarrow$	4
$\leftrightarrow$	5

p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

• Operator precedence

$\neg > \wedge > \vee > \rightarrow > \leftrightarrow$

• Logic and bits

In bit operations, 1 stands for true and 0 stands for false.

So logical operations can be applied to bit strings.

• Converse/inverse/Contrapositive

1. Quick examples

1. Original statement : $p \rightarrow q$

p: Walter cooks meth.

q: Walter breaks the law.

If Walter cooks meth, then he breaks the law

2. Converse : $q \rightarrow p$

If Walter breaks the law, then he cooks meth. (it may be false)

3. Inverse : $\neg p \rightarrow \neg q$

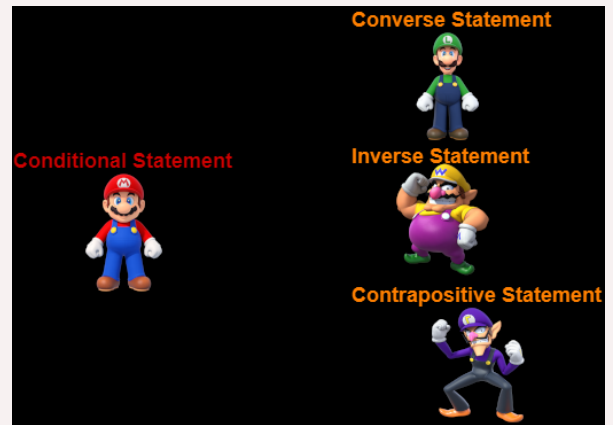
If Walter does not cook meth, then he does not break the law. (it may be false)

Walter might still break the law in some other way.

4. Contrapositive : $\neg q \rightarrow \neg p$

If Walter does not break the law, then he does not cook meth.

The original implication is equivalent to its contrapositive.



- Expressing the biconditional

- p is necessary and sufficient for q
- if p then q, and conversely
- p iff q
- p exactly when q

- Being Heisenberg is necessary and sufficient for being a famous drug dealer.
- If Mr. White is Heisenberg, then he is a famous drug dealer, and conversely.
- Mr. White is Heisenberg iff he is a famous drug dealer.
- Mr. White is Heisenberg exactly when he is a famous drug dealer.



- Expressing the implication

- | | |
|---|--|
| 1. "if p, then q" | If Mr. White cooks meth, then he breaks the law. |
| 2. "if p, q" | If Mr. White cooks meth, he breaks the law. |
| 3. "p is sufficient for q" | Mr. White cooking meth is sufficient for breaking the law. |
| 4. "q if p" | Mr. White breaks the law if he cooks meth. |
| 5. "q when p" | Mr. White breaks the law when he cooks meth. |
| 6. "a necessary condition for p is q" | A necessary condition for Mr. White cooking meth is that he breaks the law. |
| 7. "q unless ¬p" | Mr. White breaks the law unless he does not cook meth. |
| 8. "p implies q" | Mr. White cooking meth implies that he breaks the law. |
| 9. "p only if q" | Mr. White cooks meth only if he breaks the law. |
| 10. "a sufficient condition for q is p" | A sufficient condition for Mr. White breaking the law is that he cooks meth. |
| 11. "q whenever p" | Mr. White breaks the law whenever he cooks meth. |
| 12. "q is necessary for p" | Mr. White breaking the law is necessary for his cooking meth. |
| 13. "q follows from p" | Mr. White breaking the law follows from his cooking meth. |
| 14. "q provided that p" | Mr. White breaks the law, provided that he cooks meth. |

- Tautology

A tautology is a compound proposition that is always **true**

1. Quick examples

It is raining **or** it is not raining.

- Contradiction

A contradiction is a compound proposition that is always **false**.

1. Quick examples

It is raining **and** it is not raining.

- Contingency

sometimes **true**, sometimes **false**

1. Quick examples

It is raining and it is windy.

- De Morgan's law/Logical equivalence

Notation: $p \equiv q$, $p \Leftrightarrow q$, $p \leftrightarrow q$

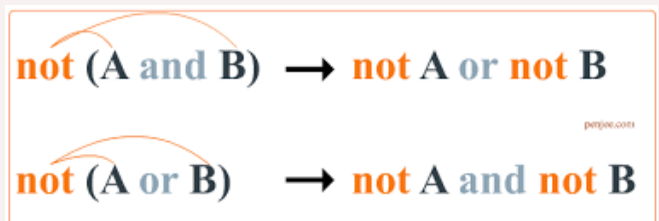
Two compound propositions are logically equivalent if $p \leftrightarrow q$ is a tautology.

De Morgan's law :

1. $\neg(p \wedge q) \equiv \neg p \vee \neg q$

2. $\neg(p \vee q) \equiv \neg p \wedge \neg q$

Proof by Contradiction:



- Identity Laws 幂等律

$$p \wedge T \equiv p, p \vee F \equiv p$$

- Domination Laws

$$p \vee T \equiv T, p \wedge F \equiv F$$

- Idempotent Laws

$$p \vee p \equiv p, p \wedge p \equiv p$$

- Double Negation

$$\neg\neg p \equiv p$$

It is raining and true $\equiv$ It is raining.

It is raining or false $\equiv$ It is raining.

It is raining or true $\equiv$ true.

It is raining and false $\equiv$ false.

It is raining or it is raining $\equiv$ It is raining.

It is raining and it is raining $\equiv$ It is raining.

It is not true that it is not raining $\equiv$ It is raining.

- Commutative

$$p \vee q \equiv q \vee p, p \wedge q \equiv q \wedge p$$

It is raining or it is windy $\equiv$ It is windy **or** it is raining.
 It is raining and it is windy $\equiv$ It is windy **and** it is raining.

- Associative Laws

$$(p \vee q) \vee r \equiv p \vee (q \vee r)$$

$$(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$$

*parentheses can be moved

(It is raining or it is windy) **or** I stay at home $\equiv$
 It is raining or (it is windy **or** I stay at home).
 (It is raining and it is windy) **and** I stay at home $\equiv$
 It is raining and (it is windy **and** I stay at home).

- Distributive Laws

$$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$$

$$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$$

It is raining or (it is windy **and** I stay at home)
 (It is raining or it is windy) **and** (it is raining or I stay at home)

- Absorption Laws

$$p \vee (p \wedge q) \equiv p$$

$$p \wedge (p \vee q) \equiv p$$

It is raining **or** (it is raining **and** it is windy) $\equiv$ It is raining.
 It is raining **and** (it is raining **or** it is windy) $\equiv$ It is raining.

- De Morgan's Laws

$$\neg(p \wedge q) \equiv \neg p \vee \neg q$$

$$\neg(p \vee q) \equiv \neg p \wedge \neg q$$

It is not the case that it is raining **and** windy. $\equiv$ Either it is not raining, or it is not windy.

It is not the case that it is raining **or** windy. $\equiv$ It is not raining and it is not windy.

- Minimal memory set

$$p \rightarrow q \equiv \neg p \vee q$$

$$p \rightarrow q \equiv \neg q \rightarrow \neg p$$

$$\neg(p \rightarrow q) \equiv p \wedge \neg q$$

$$\neg(p \wedge q) \equiv \neg p \vee \neg q$$

$$\neg(p \vee q) \equiv \neg p \wedge \neg q$$

$$p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$$

$$p \leftrightarrow q \equiv (p \wedge q) \vee (\neg p \wedge \neg q)$$

$$p \vee \neg p \equiv T, p \wedge \neg p \equiv F$$

